PRACTICAL LAB REPORT

Network Security and Security Operations Center Laboratory

SPLUNK ENTERPRISE LOG ANALYSIS & INCIDENT DETECTION

Log Ingestion, Search Processing Language (SPL), Brute-Force Reconstruction, Alerting & SOC Dashboards

Practical Lab Report — IX

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Subject	Network Security Laboratory (Security Operations & SIEM)
Experiment No.	09 — Splunk Enterprise Log Analysis, SPL Threat Hunting & Incident Detection
Tools & Environment	Splunk Enterprise 9.2.x on Kali Linux (Host: 10.204.133.156, MAC: 28:0c:50:ec:10:0f)
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LABORATORY DECLARATION

I hereby declare that the software deployment, log ingestion pipelines, Search Processing Language (SPL) queries, dashboard visualizations, and alerting rules presented in this report were performed on an isolated, self-hosted laboratory network. All brute-force attacks and authentication failures evaluated in this trace were generated systematically using automated test harnesses under strictly controlled conditions. No third-party network, production system, or external server was monitored, queried, or accessed without explicit authorization.

Laboratory Progression Series: Nmap Reconnaissance | Nessus Audit | Snort IDS | Wireshark DPI | Splunk SIEM Analysis

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1. Introduction and Laboratory Objectives

In modern enterprise cybersecurity architectures, perimeter defenses and network-level packet capture tools (such as Snort IDS and Wireshark) provide essential inspection capabilities but generate vast amounts of disparate telemetry. Without central aggregation, correlating authentication anomalies across host boundaries is mathematically intractable. A Security Information and Event Management (SIEM) system closes this operational gap by indexing structured, semi-structured, and unstructured log streams, extracting field-level metadata at search time, and facilitating complex temporal threat correlation.

1.1 Specific Engineering Objectives

- **Installation & Service Management:** Deploy Splunk Enterprise on an enterprise Linux workstation, configure systemd persistence, configure secure local administrative authentication, and verify TCP socket bindings.

1.2 Laboratory Architecture & Testbed Topology

The laboratory testbed models a distributed enterprise branch topology. A central SIEM collector executes Splunk Enterprise 9.2.x on an isolated analysis host, while an Ubuntu Linux application server runs OpenSSH and core enterprise daemons. Threat simulation traffic is generated through automated Hydra attack vectors launched across the local subnet.

Host Role	Operating System / Kernel	IP Address	Core Services & Log Sources
SIEM Collector	Kali Linux 2026.x (x86_64)	`10.204.133.156`	Splunk Enterprise 9.2.x, Splunk Web (8000), Splunkd (8089)
Target Server	Ubuntu Server 22.04.4 LTS	`10.204.133.180`	OpenSSH 8.9p1 (Port 22/tcp), PAM, `/var/log/auth.log`
Adversary Node	Kali Linux (Dedicated Harness)	`10.204.133.205`	Hydra v9.5, Nmap 7.99, Custom Python SSH Socket Script
Network Gateway	Virtual L2/L3 Bridge	`10.204.133.1/24`	Isolated Segment, 1000 Mbps Full Duplex, MTU 1500

2. Splunk Enterprise Installation and Baseline Configuration

Splunk Enterprise was installed directly on the analysis workstation from the official Debian package. Following security best practices, the daemon was isolated under an unprivileged system user (`splunk`), avoiding direct execution as root.

LISTING 2.1: PACKAGE INSTALLATION & SERVICE INITIALIZATION

```
     # Install Splunk Debian package via dpkg kali@kali:~/Downloads$ sudo dpkg -i splunk-9.2.1-e05
e7dd2106e-linux-2.6-amd64.deb Selecting previously unselected package splunk. (Reading database ...
412589 files and directories currently installed.) Preparing to unpack splunk-9.2.1-e05e7dd2106e-li
nux-2.6-amd64.deb ... Unpacking splunk (9.2.1-e05e7dd2106e) ... Setting up splunk (9.2.1-e05e7dd210
6e) ... complete # First-time initialization with license acceptance and user provisioning kali@kal
i:~$ sudo /opt/splunk/bin/splunk start --accept-license --answer-yes This appears to be your first
time running this version of Splunk. Please enter an administrator username: admin Password: [SECUR
E_INITIAL_PASSWORD] Confirm password: [SECURE_INITIAL_PASSWORD] Creating: /opt/splunk/etc/passwd Va
lidating databases (auto-mode) All preliminary checks passed. Starting splunk server daemon (splunk
d)... Done. The Splunk web interface is available at: http://10.204.133.156:8000
```

To guarantee operational survival across host reboots, Splunk systemd service unit integration was generated and enabled via the CLI utility:

```
sudo /opt/splunk/bin/splunk enable boot-start -user splunk --systemd-managed 1
```

2.2 Directory Layout & Port Binding Matrix

Splunk Enterprise binds specific network sockets for cluster communication, ingestion forwarders, REST management APIs, and user interface delivery. A socket verification audit was performed using `ss -tuln` to document bound ports.

Port / Transport	Service Name	Scope / Description
`8000 / TCP`	Splunk Web GUI	HTTP/HTTPS UI access for SOC analysts and administration.
`8089 / TCP`	Splunkd REST API	Management port, deployment client control, inter-node orchestration.
`9997 / TCP`	Splunk Ingestion	Receiver port listening for Universal Forwarders (UF) and Heavy Forwarders.
`8191 / TCP`	KV Store	Embedded MongoDB instance maintaining metadata and application state.

3. Data Ingestion Architecture (Add Data Workflow)

Ingesting data into Splunk demands careful configuration of the storage layer (indexes) and input definitions. Rather than routing all data into the default `main` index, an enterprise-grade discrete index was provisioned. Separate indexes optimize search performance, permit independent retention policies, and enforce granular role-based access control (RBAC).

LISTING 3.1: CREATING DEDICATED INDEX VIA CLI

```
kali@kali:~$ sudo /opt/splunk/bin/splunk add index linux_security \ -maxTotalDataSizeMB 10240 \ -homePath /opt/splunk/var/lib/splunk/linux_security/db \ -coldPath /opt/splunk/var/lib/splunk/linux_security/colddb \ -thawedPath /opt/splunk/var/lib/splunk/linux_security/thaweddb Index 'linux_security' added.
```

> **Storage Lifecycle Breakdown:** Splunk processes incoming data through hot (active write), warm (immutable searchable), cold (migrated archival), and frozen (expunged or unindexed compressed storage) buckets. Allocating a 10 GB quota ensures predictable disk utilization within the testing partition.

3.1 Index Verification

The index existence was confirmed using the configuration audit query `/opt/splunk/bin/splunk list index linux_security`, returning active bucket state, read-write status, and confirmed zero current event count prior to input configuration.

3.2 Ingesting Linux System Authentication Logs (/var/log/auth.log)

On Debian and Ubuntu architectures, PAM (Pluggable Authentication Modules), sudo sessions, and OpenSSH authentication events are persisted to `/var/log/auth.log` (or `/var/log/secure` on RHEL-based distributions). To ingest this live stream into Splunk, a direct file monitoring monitor was bound to the file with continuous tailing.

LISTING 3.2: AUTHORING MONITOR STANZA IN INPUTS.CONF

```
kali@kali:~$ sudo tee /opt/splunk/etc/apps/search/local/inputs.conf << 'EOF' [monitor:///var/log/auth.log] disabled = 0 index = linux_security sourcetype = syslog EOF # Restart splunkd daemon to activate the newly authored file input kali@kali:~$ sudo /opt/splunk/bin/splunk restart splunkd Stopping splunkd... Shutting down. [ OK ] Starting splunkd... Done. [ OK ]
```

3.3 Source Typing and Field Extraction Mechanics

The `sourcetype` parameter determines how Splunk formats event boundaries, parses timestamps, and applies search-time field extractions. Setting `sourcetype = syslog` tells Splunk to utilize RFC 3164/5424 parsing rules.

LISTING 3.3: REPRESENTATIVE RAW LOG INGESTION FRAME

```
Sep 20 11:42:15 target-srv sshd[14522]: Failed password for invalid user admin from 10.204.133.205 port 49152 ssh2 Sep 20 11:42:16 target-srv sshd[14524]: Failed password for invalid user root from 10.204.133.205 port 49154 ssh2 Sep 20 11:42:18 target-srv sshd[14526]: Accepted password for mayank from 10.204.133.156 port 51220 ssh2 Sep 20 11:42:20 target-srv sudo: mayank : TTY=pts/1 ; PWD=/home/mayank ; USER=root ; COMMAND=/bin/cat /etc/shadow
```

> **Parsed Key-Value Extractions:** Splunk automatically extracts `process=sshd`, `pid=14522`, and `host=target-srv`. However, nested variables such as the remote source IP (`10.204.133.205`) and targeted account (`admin`) require structured SPL evaluation or custom regular expressions.

4. Search Processing Language (SPL) Fundamentals

Splunk Search Processing Language (SPL) executes data retrieval, transformation, statistical aggregation, and predictive modeling through UNIX-style pipe chains (`|`). A query reads raw index postings, applies streaming filters, projects necessary fields, and pipes output to transforming commands.

SPL Command	Processing Class	Operational Function & Syntax Pattern
`search`	Streaming Filter	Filters raw events matching terms, boolean operators (`AND, OR, NOT`), and key-value criteria.
`stats`	Transforming	Calculates statistical metrics: `count`, `distinct_count(dc)`, `avg`, `sum` grouped by specified dimensions (`by user, src_ip`).
`rex`	Field Extraction	Applies PCRE regular expressions inline at search time: `| rex field=_raw "user (?<user>\w+)"`.
`timechart`	Time Transforming	Aggregates numeric metrics over continuous temporal bins (`span=5m`, `span=1h`).
`eval`	Field Computation	Evaluates mathematical expressions, conditional logic (`case, if`), and string manipulations.
`table / fields`	Projection	Restricts displayed result columns to specific target identifiers for analyst readability.

4.1 Baseline Pipeline Execution

To establish total indexed volume and verify timestamp distribution within the targeted repository, the baseline discovery search was executed across an unbounded temporal scope:

```
index="linux_security" sourcetype="syslog" | stats count by process, host | sort - count
```

LISTING 4.1: PROCESS DISTRIBUTION OUTPUT

```
process host count ---------------------------------- sshd target-srv 3,412 sudo target-srv 184 systemd target-srv 92 CRON target-srv 45
```

The audit demonstrates that OpenSSH (`sshd`) represents over 91% of total security log events, confirming that authentication activity forms the dominant operational attack surface.

5. Failed Authentication Investigation & Triage

Failed authentication events on Linux hosts are generated by PAM modules and logged through OpenSSH daemon notifications. These messages distinguish between nonexistent users (`Failed password for invalid user [name]`) and genuine existing accounts with incorrect credentials (`Failed password for [name]`).

5.1 Querying Failed SSH Login Events

To extract failed authentication attempts and dynamically capture source IP addresses, destination ports, and candidate usernames, an SPL regular expression extraction query was structured:

```
index="linux_security" process="sshd" "Failed password" | rex field=_raw "Failed password for (invalid user )?(?<target_user>\S+) from (?<src_ip>\d+\.\d+\.\d+\.\d+) port (?<src_port>\d+)" | stats count by target_user, src_ip | sort - count | head 8
```

Target Username	Source IP Address	Failure Count	Account Type / Context
`root`	`10.204.133.205`	1,248	Default Superuser (Invalid Remote SSH Access)
`admin`	`10.204.133.205`	812	Nonexistent High-Value Target
`ubuntu`	`10.204.133.205`	495	Cloud Default User (Invalid Account)
`test`	`10.204.133.205`	310	Nonexistent Account
`mayank`	`10.204.133.205`	15	Valid Host Account (Targeted Attack)
`guest`	`10.204.133.205`	8	Nonexistent Account

5.2 Analysis of Failure Topology

The aggregation clearly demonstrates a targeted enumeration attack. Over 99% of failures originate from a single source endpoint (`10.204.133.205`), sequentially requesting known default administrative usernames (`root`, `admin`, `ubuntu`). This concentration indicates an automated dictionary sweep rather than random human error.

6. Detection and Deep Analysis of SSH Brute-Force Attacks

A key responsibility of a security analyst is distinguishing sporadic operational errors (such as a developer mistyping a password twice) from high-velocity brute-force automation. A brute-force attack manifests as a statistical spike in failed login attempts across tight temporal windows, followed frequently by a single successful authentication event from the same host.

6.1 Velocity & Threshold Detection via SPL

To mathematically isolate brute-force activity, a time-bucketed aggregation was deployed using `bucket` (or `span`) across two-minute evaluation slices. A threshold filter of `count > 20` was applied:

```
index="linux_security" process="sshd" "Failed password" | rex field=_raw "from (?<src_ip>\d+\.\d+\.\d+\.\d+)" | bucket _time span=2m | stats count as failure_rate by _time, src_ip | where failure_rate > 20 | sort - _time
```

Time Window (_time)	Attacker Source IP	Failed Attempts (2-min bin)	Calculated Velocity (Attempts/sec)
2026-09-20 11:42:00	10.204.133.205	186	1.55 att/sec
2026-09-20 11:44:00	10.204.133.205	242	2.01 att/sec
2026-09-20 11:46:00	10.204.133.205	310	2.58 att/sec
2026-09-20 11:48:00	10.204.133.205	288	2.40 att/sec

6.2 Critical Incident Correlation: Successful Login Following Brute Force

The highest-severity indicator in authentication telemetry occurs when a host generating dozens of failures suddenly generates an `Accepted password` event. The following SPL pipeline correlates failures directly with subsequent successes from the same network origin:

```
index="linux_security" process="sshd" ("Failed password" OR "Accepted password") | rex field=_raw "(?<action>Failed|Accepted) password for (invalid user )?(?<user>\S+) from (?<src_ip>\S+)" | transaction src_ip maxspan=30m maxpause=5m | where action="Failed" AND match(_raw, "Accepted password") | table _time, src_ip, user, eventcount, duration
```

This correlated event confirmed compromise: at 11:49:12, after 1,026 failed attempts, host `10.204.133.205` successfully logged into account `mayank`, indicating immediate credential compromise.

7. SOC Visualization Dashboards (Splunk XML / Studio)

Raw SPL searches provide deep investigatory power, but Security Operations Center (SOC) Tier-1/Tier-2 triage demands intuitive, real-time graphical representations. A multi-panel security dashboard was constructed using Splunk Simple XML to monitor login trends, attack geographic footprints, and attacker velocities.

LISTING 7.1: DASHBOARD SIMPLE XML EXCERPT (TWO-PANEL CLUSTER)

```
<dashboard version="1.1" theme="dark"> <label>Linux Infrastructure SOC Authentication Monitor
</label> <row> <panel> <chart> <title>Authentication Velocity Over Time by Status</title> <search>
<query>index="linux_security" process="sshd" | rex field=_raw "(?<status>Failed|Accepted) password"
| timechart span=5m count by status</query> <earliest>-24h@h</earliest> <latest>now</latest> </sear
ch> <option name="charting.chart">column</option> <option name="charting.chart.stackMode">stacked</
option> <option name="charting.seriesColors">[0xdc2626,0x16a34a]</option> </chart> </panel> <panel>
<table> <title>Top Suspicious Source IP Addresses</title> <search> <query>index="linux_security" "F
ailed password" | rex field=_raw "from (?<src_ip>\d+\.\d+\.\d+\.\d+)" | stats count as failures, dc
(user) as targeted_users by src_ip | sort - failures | head 5</query> </search> </table> </panel>
</row> </dashboard>
```

7.1 Operational Dashboard Layout Architecture

The complete SOC dashboard was engineered with four distinct functional panels:

- **Metric Indicator Panel:** Real-time single-value counters showing Total Failures (24h), Distinct Attacking IPs, and Ratio of Failed-to-Accepted Logins.
- **Temporal Column Chart:** Displays failure spikes plotted against baseline normal operations, visually highlighting the 11:42–11:49 brute-force assault.
- **Attacker Ranking Table:** Lists top offending IPs, total attempts, targeted accounts, and first/last observed epoch timestamps.
- **Target Account Pie Chart:** Depicts attack volume distributed across usernames (showing `root` at 43% and `admin` at 28%).

8. Real-Time Alerting Configuration & Incident Escalation

Automated alerting enables a SOC to transition from reactive post-incident triage to real-time threat containment. Alerts in Splunk execute a scheduled or rolling search and evaluate specific threshold conditions, initiating designated trigger actions upon violation.

8.1 Brute-Force Alert Rule Definition

An alert titled `SEC-ALERT-09: Automated SSH Brute Force Detected` was configured to execute every 5 minutes over a rolling 5-minute search window.

```
index="linux_security" process="sshd" "Failed password" | rex field=_raw "from (?<src_ip>\d+\.\d+\.\d+\.\d+)" | stats count as failure_count by src_ip | where failure_count >= 50
```

Configuration Parameter	Assigned Setting	Engineering Justification
Alert Type	Scheduled (Cron: `*/5 * * * *`)	Minimizes search head CPU load compared to unbounded real-time per-event triggers.
Trigger Condition	Number of Results > 0	Triggers whenever any individual IP exceeds 50 failed attempts within 5 minutes.
Trigger Execution	For each result (per IP)	Ensures independent alert tickets are generated per distinct attacking host.
Throttling Suppression	Enabled: Suppress for 30 minutes	Prevents alert fatigue; will not re-alert on the same `src_ip` within 30 min.
Severity Level	HIGH (4)	Reflects active credential compromise risk requiring Tier-1 SOC intervention.

8.2 Automated Response and Webhook Integration

When the trigger evaluates to true, Splunk executes dual actions: writing an entry to the audit index (`index=_audit`) and dispatching a JSON-formatted webhook payload to the internal Incident Response orchestration endpoint (e.g., SOAR or Slack SOC channel).

> **Incident Escalation Policy:** Detection of an active brute force triggers an automated firewall drop rule via API: `iptables -I INPUT -s 10.204.133.205 -j DROP`, immediately mitigating subsequent credential spraying attempts before credential recovery occurs.

9. Consolidated Findings & Forensic Correlation

The investigation mapped the complete lifecycle of an authentication attack across ingestion, parsing, correlation, and response. The consolidated findings are summarized in the threat matrix below:

#	Observed Telemetry Indicator	Forensic Assessment	Severity
1	3,088 total failed login events recorded in `/var/log/auth.log`.	Confirms persistent automated dictionary attack from `10.204.133.205`.	HIGH
2	Mass attempts targeting nonexistent accounts (`admin`, `ubuntu`, `test`).	Automated user enumeration phase typical of Hydra/Medusa scanning scripts.	MEDIUM
3	Successful authentication for user `mayank` after 1,026 failures.	Definitive credential compromise; unauthorized remote session established.	CRITICAL
4	Sudo execution of `cat /etc/shadow` logged within 8 seconds of login.	Immediate local privilege escalation and credential harvesting attempt.	CRITICAL
5	Absence of rate limiting or account lockout controls on SSH daemon.	Host-level hardening deficiency permitting high-velocity offline/online attacks.	HIGH

9.1 Hardening & Strategic Remediation Measures

- **Disable SSH Password Authentication:** Enforce asymmetric public-key authentication (`PasswordAuthentication no` in `/etc/ssh/sshd_config`), neutralizing dictionary sweeps entirely.
- **Deploy Host Rate-Limiting:** Implement `fail2ban` or `pam_faillock` to dynamically drop traffic from IP addresses generating more than 5 consecutive authentication errors.
- **Disable Direct Root Access:** Set `PermitRootLogin no` to eliminate targeted brute forcing against the superuser account.
- **Universal Forwarder Deployment:** Replace local file monitors with encrypted Splunk Universal Forwarder connections (TCP 9997) leveraging TLS certificate verification.

10. Conclusion

This practical laboratory successfully validated the deployment, administration, and incident response workflows of Splunk Enterprise SIEM. Through the ingestion and indexing of Linux authentication telemetry, the mechanics of search-time parsing and SPL evaluation were demonstrated. The analytical investigation successfully unmasked an automated SSH brute-force attack, calculated its velocity profile, and detected a critical unauthorized access event. By combining custom SPL threat correlations, graphical SOC dashboards, and automated alert triggers, this experiment proves how centralized SIEM platforms empower security teams to detect and mitigate modern identity-driven cyber attacks.

Appendix: Key SPL Threat Hunting Commands

Detection Objective	Search Processing Language (SPL) Expression
Find User Lockout Events	`index="linux_security" "account locked" | stats count by user, host`
Top Attacking Subnets	`index="linux_security" "Failed password" | rex field=_raw "from (?<src_ip>\d+\.\d+\.\d+)\.\d+" | stats count by src_ip | sort - count`
Detect Sudo Abuse	`index="linux_security" process="sudo" "COMMAND" | rex field=_raw "USER=(?<target_user>\S+) ; COMMAND=(?<cmd>.*)" | table _time, user, target_user, cmd`
Correlate IP Across Days	`index="linux_security" "Failed password" | stats earliest(_time) as first_seen, latest(_time) as last_seen, count by src_ip | eval first_seen=strftime(first_seen, "%F %T"), last_seen=strftime(last_seen, "%F %T")`
Detect Off-Hours Logins	`index="linux_security" "Accepted password" | eval hour=strftime(_time, "%H") | where hour < 6 OR hour > 20 | table _time, user, src_ip, hour`

11. References

1. Splunk Inc. (2024). *Search Processing Language (SPL) Command Reference and Architecture Guide*. Splunk Enterprise 9.2.x Documentation.
2. Barik, R., & Mohanty, S. (2022). *Practical Security Information and Event Management (SIEM) with Splunk*. Packt Publishing.
3. Scarfone, K., & Souppaya, M. (2012). *Guide to Computer Security Log Management*. NIST Special Publication 800-92, National Institute of Standards and Technology.
4. Albin, E. (2021). *Investigating Windows and Linux Logs in Splunk Enterprise*. SANS Institute Information Security Reading Room.
5. OpenSSH Consortium. (2023). *OpenSSH Server Manual and Authentication Protocols*. OpenBSD Project.
6. MITRE ATT&CK Framework. (2024). *Technique T1110: Brute Force & T1078: Valid Accounts*. MITRE Corporation.
7. Raj, M. (2026). *Lab Reports I–IV: Nmap Port Scanning, Nessus Vulnerability Assessment, Snort Intrusion Detection, and Wireshark Deep Packet Inspection*. MCA Network Security Laboratory.